GS EXTERIOR EXPERTS · DECK DIVISION · MATERIAL TAKE-OFF + BUILD SET · FOR DECKS & DOCKS

281 S Jefferson Ave · Decks + Roof Covers

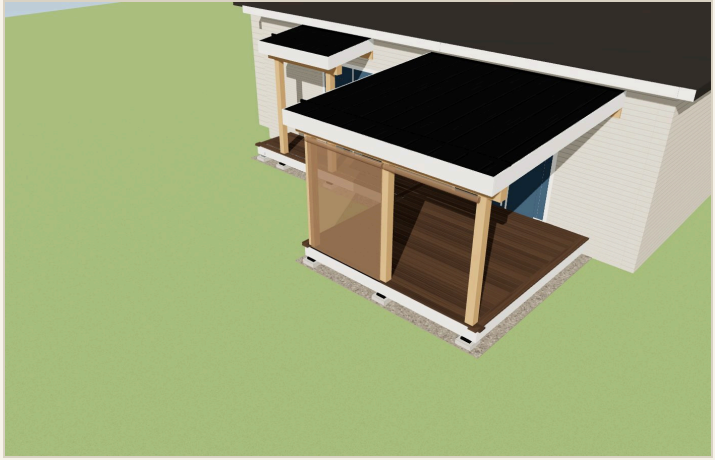
281 S Jefferson Ave, Louisville · two decks (12' x 12' + 14'8" x 3'8" walkway) on CEMCO steel and Diamond Piers · TimberTech Landmark American Walnut, picture frame + dividers, no butt joints · two lean-to covers, black standing seam, Hardie soffit / fascia / rakes · blinds kept · October 16, 2026

Account GSEXT1 · ship to 281 S Jefferson Ave, Louisville, CO 80027 · please quote every line; call Jade Pender 303-550-9558 with substitutions.

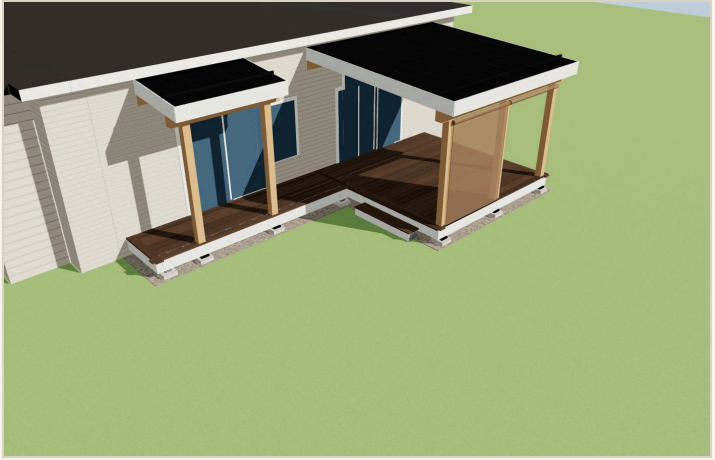
SQUARE FEET	FOOTINGS	ORDER LINES	HEIGHT
197.8	**7**	**78**	**1'-3"**

THE JOB

Deck	197.8 SF · 1'-3" above grade · 2 zones, 26'-8" along the house, 12'-0" at the deepest
Joists	7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC
Rims	single-ply 7-1/4" steel track rims and ledger track on a steel angle
Beams	FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,616 lb)
Footings / posts	2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam on DP-50/50
Decking	TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc
Rail	none
Stairs	main deck side, toward the walkway: 2 risers @ 7.50", 1 treads, 6 stringers, 5'-0" wide



Isometric from the stair corner



Isometric from the open corner

MATERIAL TAKE-OFF NET = counted off the drawings · ORDER = what to pull and ship

Lumber

NET	ORDER	UNIT	ITEM	WHY
347	**364**	LF	CEMCO 7-1/4" steel C-joist, 16 ga (54 mil) G90 — joists, PF joists, blocking	+5% (cut to length by the supplier where possible) — 12" OC
77	**82**	LF	CEMCO 7-1/4" steel track, 16 ga G90 — rims (single ply, caps the joist ends) + ledger track	+5% (51 LF rim + 27 LF ledger)
27	**30**	LF	Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track	+10%
51	**54**	LF	CEMCO 7-1/4" steel C, 14 ga (68 mil) G90 — boxed beams (2 C toe-to-toe)	+5% (FLUSH (2)2x8 SYP W 14.2' · FLUSH (2)2x8 SYP M 11.5')
5	**5**	ea	Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)	exact (one per shared post, (2) 1/2" thru-bolts + #12 screws into the beam)
2	**2**	ea	Steel post stub 3" x 3" x 11 ga HSS, hot-dip galvanized, cap plate welded — cut to height	exact (one per deck-only pier; 6" clear)
2	**2**	ea	Pier-head post bracket for the Diamond Pier bolt (steel post)	exact

Footings

NET	ORDER	UNIT	ITEM	WHY
7	**7**	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
1	**1**	ea	Upgrade to Diamond Pier DP-75/63 at the shared main cover posts (roof + deck on one pier)	exact (20'-8": 3,317 lb vs 3,300 lb)

Hardware

NET	ORDER	UNIT	ITEM	WHY
56	**62**	ea	Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam	+6 (2 per joist end where it frames into a beam or ledger)
3	**3**	box	#12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct	1463 needed (clips, track, boxed beams, post brackets)

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer)	40 needed — 2 rows staggered 16" OC
8	**8**	ea	Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent	exact (2 near each end of every ledger)
2	**2**	ea	Cold-galvanizing spray (cut ends, field drilling)	lot

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	**1**	roll	Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)	27 LF of house wall with laps — nothing on a privacy wall
3	**3**	ea	Aluminum L-flashing 10' (over ledger + side rims)	house walls only
1	**1**	roll	6-mil poly sheeting 10' x 100' — under the decks	218 SF with laps
1.3	**1.5**	CY	3/4" washed gravel, 2" over the poly — under the decks (delivered)	218 SF x 2"

Decking

NET	ORDER	UNIT	ITEM	WHY
25	**27**	ea	TimberTech Landmark American Walnut 1x6x12 Grooved	+2 (25 x 12' · 7 x 16' · panel 1 7 rows · panel 2 25 rows · every row one piece)
7	**9**	ea	TimberTech Landmark American Walnut 1x6x16 Grooved	+2 (25 x 12' · 7 x 16' · panel 1 7 rows · panel 2 25 rows · every row one piece)
4	**4**	ea	TimberTech Landmark American Walnut 1x6x16 Square Edge — borders & dividers	exact (border 14'-8",12'-0" 12'-1" · border 26'-8",0" 12'-1" · breaker DW/M 12'-1" · border 14'-8",3'-8" 8'-4 15/16")
1	**2**	ea	TimberTech Landmark American Walnut 1x6x20 Square Edge — borders & dividers	+1 (border 0",3'-8" 14'-9" + border 0",0" 3'-9")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)	437 needed · 500 supplied
1	**1**	box	Starborn Cap-Tor xd 2-3/4" Epoxy American Walnut 350 ct	284 screws needed (first row 28 · border 116 · fascia 92 · stairs 48)

Fascia

NET	ORDER	UNIT	ITEM	WHY
5	**6**	ea	Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia)	+1 (left fascia (W) 3'-8 1/2" · right fascia (W) 3'-8 1/2" · front fascia (W) 12'-0" · front fascia (W) 2'-9" · left fascia (M) 12'-0" · left fascia (M) 1/2" · right fascia (M) 12'-0" · right fascia (M) 1/2" · front fascia (M) 12'-0" · front fascia (M) 1")
1	**1**	box	Color-matched trim screws for HardieTrim, 100 ct (Arctic White)	40 color-matched screws, pre-drilled, 2 per joist line

NET	ORDER	UNIT	ITEM	WHY
1	**1**	kit	ColorPlus touch-up kit, Arctic White	cut ends

Stairs

NET	ORDER	UNIT	ITEM	WHY
1	**2**	ea	TimberTech Landmark American Walnut 1x6x12 Square Edge — treads (main deck side, toward the walkway stair)	+1 (1 treads x 2 boards @ 5'-0", 2 per board)
1	**1**	ea	TimberTech Landmark American Walnut riser board 7-1/4" x 12' (main deck side, toward the walkway stair)	exact (2 risers @ 5'-0")
2	**2**	ea	Steel stair stringer, galvanized, pre-formed to the riser / tread	exact

Porch cover

NET	ORDER	UNIT	ITEM	WHY
3	**3**	ea	6x6x10 Douglas fir #1 post (cover) — Main cover	exact (3 posts on the beam line, field-cut)
2	**2**	ea	2x10x16 #1 SYP — cover beam (2 ply) — Main cover	+0 ((2)2x10 x 12.0' at the posts)
1	**1**	ea	2x10x16 #1 SYP — cover ledger on the house — Main cover	exact
10	**11**	ea	2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover	+1 cull (10 rafters, 13.2' with the overhang)
3	**4**	ea	2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover	+1 (38 LF)
6	**6**	sheet	7/16" OSB roof sheathing 4x8 — Main cover	170 SF + 10% cuts
170	**178**	SF	Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 13.2' (Black, PVDF), 10 panels — Main cover	10 panels x 13.2' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier
1	**1**	roll	High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover	170 SF — full coverage under metal (high-temp)
2	**3**	ea	Standing seam eave / drip trim 10', color-matched — Main cover	12 LF low eave + 1
3	**4**	ea	Standing seam gable / rake trim 10', color-matched — Main cover	two rakes x 13.2' + 1
2	**3**	ea	Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover	12 LF at the house — counterflash into the siding / brick
3	**3**	ea	Foam + metal closure strips at eave and headwall, 10' — Main cover	eave + headwall closures
1	**1**	box	Panel clips + pancake-head screws, box of 250 — Main cover	90 clips @ 18" OC on each seam
1	**1**	box	Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover	trims, closures, snow bar
3	**3**	ea	Butyl tape / polyurethane sealant, per roll or tube — Main cover	headwall, eave, rake laps
12	**14**	LF	Snow retention — clamp-on bar across the low eave over the deck (Black) — Main cover	12' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave
5	**5**	panel	Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Main cover	170 SF + 10% cuts

NET	ORDER	UNIT	ITEM	WHY
3	**3**	lb	Stainless ring-shank siding nails 2", per lb — Main cover	stainless ring-shank, 6" OC at every rafter
4	**5**	ea	Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Main cover	+1 (38 LF: front 12' + two rakes 13.2')
1	**1**	box	Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Main cover	color-matched trim screws
3	**3**	ea	Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover	exact
3	**3**	ea	Simpson AC6Z post cap, 6x6 to (2)2x10 beam — Main cover	exact
10	**10**	ea	Simpson H2.5AZ rafter tie — Main cover	one per rafter at the beam
19	**19**	ea	FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Main cover	12 LF x 2 rows @ 16"
2	**2**	ea	6x6x10 Douglas fir #1 post (cover) — Walkway cover	exact (2 posts on the beam line, field-cut)
2	**2**	ea	2x10x16 #1 SYP — cover beam (2 ply) — Walkway cover	+0 ((2)2x10 x 6.3' at the posts)
1	**1**	ea	2x10x16 #1 SYP — cover ledger on the house — Walkway cover	exact
5	**6**	ea	2x8x8 #1 SYP — cover rafters @ 16" OC — Walkway cover	+1 cull (5 rafters, 4.9' with the overhang)
2	**3**	ea	2x8x16 #1 SYP — cover fascia / sub-fascia — Walkway cover	+1 (16 LF)
2	**2**	sheet	7/16" OSB roof sheathing 4x8 — Walkway cover	35 SF + 10% cuts
35	**37**	SF	Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 4.9' (Black, PVDF), 5 panels — Walkway cover	5 panels x 4.9' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier
1	**1**	roll	High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Walkway cover	35 SF — full coverage under metal (high-temp)
1	**2**	ea	Standing seam eave / drip trim 10', color-matched — Walkway cover	6 LF low eave + 1
1	**2**	ea	Standing seam gable / rake trim 10', color-matched — Walkway cover	two rakes x 4.9' + 1
1	**2**	ea	Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Walkway cover	6 LF at the house — counterflash into the siding / brick
2	**2**	ea	Foam + metal closure strips at eave and headwall, 10' — Walkway cover	eave + headwall closures
1	**1**	box	Panel clips + pancake-head screws, box of 250 — Walkway cover	20 clips @ 18" OC on each seam
1	**1**	box	Color-matched #10 woodgrip trim screws w/ washers, 250 — Walkway cover	trims, closures, snow bar
3	**3**	ea	Butyl tape / polyurethane sealant, per roll or tube — Walkway cover	headwall, eave, rake laps
6	**8**	LF	Snow retention — clamp-on bar across the low eave over the deck (Black) — Walkway cover	6' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave

NET	ORDER	UNIT	ITEM	WHY
1	**1**	panel	Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Walkway cover	35 SF + 10% cuts
1	**1**	lb	Stainless ring-shank siding nails 2", per lb — Walkway cover	stainless ring-shank, 6" OC at every rafter
2	**3**	ea	Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Walkway cover	+1 (16 LF: front 6' + two rakes 4.9')
1	**1**	box	Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Walkway cover	color-matched trim screws
2	**2**	ea	Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover	exact
2	**2**	ea	Simpson AC6Z post cap, 6x6 to (2)2x10 beam — Walkway cover	exact
5	**5**	ea	Simpson H2.5AZ rafter tie — Walkway cover	one per rafter at the beam
10	**10**	ea	FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Walkway cover	6 LF x 2 rows @ 16"

FASTENER & CONNECTOR SCHEDULE

Ledger	7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
Field	CONCEALoc at every joist in every gap (437)
Face screws	Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board
Steel frame	7-1/4" C-joists 16 ga @ 12" OC · single-ply track rims · boxed C beams · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed

CUT LIST

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists (W)	7-1/4" C	3'-3 1/2"	13	@ 12" OC, crown up
Picture-frame joists (W)	7-1/4" C	3'-3 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (W)	7-1/4" C	3'-3 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Ledger (W)	7-1/4" C	14'-8"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (W)	7-1/4" C	10 1/2"	14	1 row at 1'-10" from the wall
Field joists (M)	7-1/4" C	11'-7 1/2"	11	@ 12" OC, crown up
Picture-frame joists (M)	7-1/4" C	11'-7 1/2"	2	centre 5 1/4" from each frame face
Divider doublers (M)	7-1/4" C	11'-7 1/2"	1	sister a joist on the divider line so both board edges bear

MEMBER	STOCK	LENGTH	QTY	NOTE
Side rim plies (M)	7-1/4" C	11'-7 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Ledger (M)	7-1/4" C	12'-0"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (M)	7-1/4" C	10 1/2"	12	1 row at 6'-0" from the wall
Field boards panel 1	Landmark American Walnut	14'-0"	6	one board wall to border, no joints
Field boards panel 1 (rip)	Landmark American Walnut	14'-0"	1	one board wall to border, no joints; ripped to fit the border / the house
Field boards panel 2	Landmark American Walnut	11'-3"	18	one board wall to border, no joints
Field boards panel 2	Landmark American Walnut	11'-4"	6	one board wall to border, no joints
Field boards panel 2 (rip)	Landmark American Walnut	11'-4"	1	one board wall to border, no joints; ripped to fit the border / the house
Stair stringers (main deck side, toward the walkway)	steel stringer	2'-7"	6	2 risers @ 7.500" · 1 treads @ 11.66" · stringers 12" OC
FLUSH (2)2x8 SYP W beam	7-1/4" C	14'-2"	2	y 3'-6 1/2"; posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16"
Posts under FLUSH (2)2x8 SYP W	3x3 HSS	6"	4	field-measure each
FLUSH (2)2x8 SYP M beam	7-1/4" C	11'-6"	2	y 11'-10 1/2"; posts at 15'-2" / 20'-8" / 26'-2"
Posts under FLUSH (2)2x8 SYP M	3x3 HSS	6"	3	field-measure each
Border 26'-8",0"	Landmark American Walnut square	12'-1"	1	mitre at the corners; dividers on the doubled joist
Border 14'-8",12'-0"	Landmark American Walnut square	12'-1"	1	mitre at the corners; dividers on the doubled joist
Border 14'-8",3'-8"	Landmark American Walnut square	8'-4 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 0",3'-8"	Landmark American Walnut square	14'-9"	1	mitre at the corners; dividers on the doubled joist
Border 0",0"	Landmark American Walnut square	3'-9"	1	mitre at the corners; dividers on the doubled joist
Breaker dw/m	Landmark American Walnut square	12'-1"	1	mitre at the corners; dividers on the doubled joist
Stair treads (main deck side, toward the walkway)	Landmark American Walnut square	5'-0"	2	2 boards per tread, nosing 3/4" over the riser
Stair risers (main deck side, toward the walkway)	riser board	5'-0"	2	rip to 7.50" less tread thickness
Left fascia (w)	fascia	3'-8 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (w)	fascia	3'-8 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house

MEMBER	STOCK	LENGTH	QTY	NOTE
Front fascia (w)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (w)	fascia	2'-9"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (m)	fascia	1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (m)	fascia	1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (m)	fascia	1"	1	flush under the board nose; full board at the front corners, short piece at the house

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

PROJECT

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel, TimberTech Landmark American Walnut)
281 S Jefferson Ave, Louisville CO 80027
197.8 SF · 1'-3" above grade · 2 zones

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; ground snow per the ASCE hazard tool (Louisville criteria) — 30 psf used here (city sheet, non-reducible); Boulder County layer shows 40 psf: engineer confirms
Dead load	10 psf deck · 15 psf porch cover (standing seam steel on sheathing, Hardie soffit; 12 psf actual, 15 used) carried on the deck posts
Roof snow (cover)	30 psf flat-roof design snow, sliding snow off the house roof + drift at the house wall per ASCE 7 Ch. 7
Design total	50 psf on the deck · cover posts add roof snow + dead to the deck frame and footings (engineered)
Wind	145 mph Vult (3-second gust), Exposure C (Louisville 2024 design criteria), Risk Cat. II
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2024 IRC / IBC as adopted and amended by the City of Louisville (Ord. 1907, Series 2025); IRC R507 decks with LMC 15.05.140; IBC Appendix I patio covers; ASCE 7-22 loads; Louisville Fire Hardening provisions; cold-formed steel frame per AISI S240, engineered
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam
Beams	FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,616
Joists	7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC
Rims	single-ply 7-1/4" steel track rims and ledger track on a steel angle
Ledger	7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
Decking	TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc
Rail	none

SHEET INDEX

G-001	General notes · design criteria
C-100	Site plan
A-001	Isometric · renders
S-100	Foundation plan
S-101	Framing plan
A-101	Decking plan
A-301	Section + front elevation
A-401	Roof covers — plan + section
D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

Rear / front lot line: deck edge 12.0' from the line, setback per the PUD...
Side lot line: deck edge 9.8' from the line, setback per the PUD —...

No scale

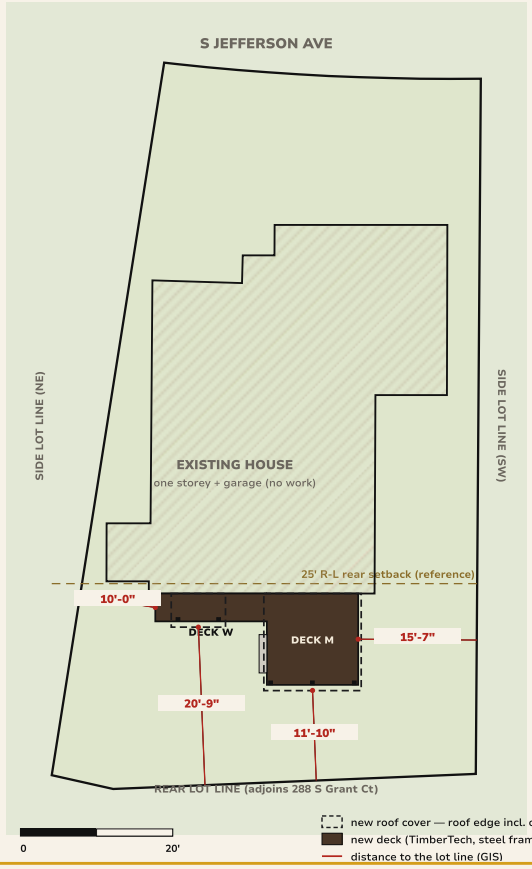
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

C-100 SITE PLAN

Parcel, existing house, new decks and roof covers, distances to the lot lines, coverage

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0





PARCEL

281 S Jefferson Ave, Louisville CO 80027
Parcel 157517119033 · Account R0092293
Lot 33, Block 10, Dutch Creek (plat rec. 00492001, 1982)
Section 17, T1S, R69W
Zoning: R-L within the Dutch Creek PUD (approved 12/15/1981)
Lot 4,678 SF · flood zone X
House 1982 one-storey ranch: 920 SF main floor + 400 SF garage (assessor)

LOT COVERAGE

Lot area	4,678 SF
House + garage (assessor footprint)	1,320 SF
Existing covers to the posts (2024 aerial, est.)	~142 SF (main ~12' x 10'-4", walkway ~5'-4" x 3'-5")
Existing coverage	~1,462 SF · 31.3%
Main cover, new (to the posts) 12'-0" x 11'-9"	141 SF
Walkway cover, new (to the posts) 5'-4" x 3'-5"	18 SF
Proposed coverage	~1,479 SF · 31.6% (+~17 SF)
Uncovered decks	not counted (under 30" above grade)

SETBACKS

Decks are 30" or less above grade — not held to the principal setbacks (city deck handout)
Roof covers: part of the principal structure — they replace the two existing covers in place; the 2024 aerial shows the existing main roof ~1'-5" shallower: field-verify the existing post line
R-L reference setbacks (LMC 17.12.040) — the Dutch Creek PUD plan governs; Planning to confirm
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the two decks, the two roof covers and the box step; haul off
· Diamond Pier foundations (no excavation spoils), 6-mil poly + 2" gravel under the decks
· Stair lands on grade (no pad)
· Roof drainage: covers shed to grade at the low eave, away from the house
· Call 811 before the piers; protect the lawn and the owner's blinds

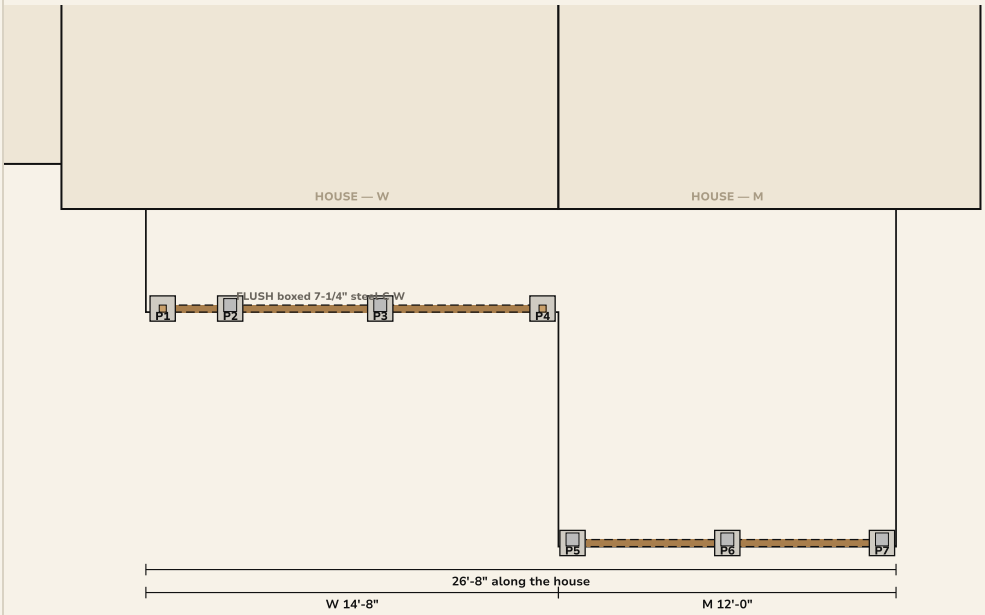
Boulder County GIS parcel + 2024 Pictometry roof outline, Louisville GIS zoning / PUD layers. GIS lines are not a survey — an ILC governs if Planning requires one.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	7 3/16"	3'-6 1/2"	W	132 lb
P2	3'-0"	3'-6 1/2"	W	330 lb
P3	8'-4"	3'-6 1/2"	W	474 lb
P4	14'-1 3/16"	3'-6 1/2"	W	273 lb
P5	15'-2"	11'-10 1/2"	M	881 lb
P6	20'-8"	11'-10 1/2"	M	1,616 lb
P7	26'-2"	11'-10 1/2"	M	881 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,616 lb

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FRAMING NOTES

7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC

single-ply 7-1/4" steel track rims and ledger track on a steel angle

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board

blocking one row over the beam, staggered

FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4"

Connectors:

Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam: 56

Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2

Cold-galvanizing spray (cut ends, field drilling): 2

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-101 Decking plan

A-101 **DECKING PLAN**

Every board, border and divider as it lays; rail posts shown for the pockets

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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DECKING

Field: TimberTech Landmark American Walnut, 5.5" x 1", 3/16" gaps

Border & dividers: Landmark American Walnut

Boards run parallel to the house; edge overhang 1" over the fascia

CONCEALoc at every joist in every gap (437)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker — every board one piece wall to border, no butt

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

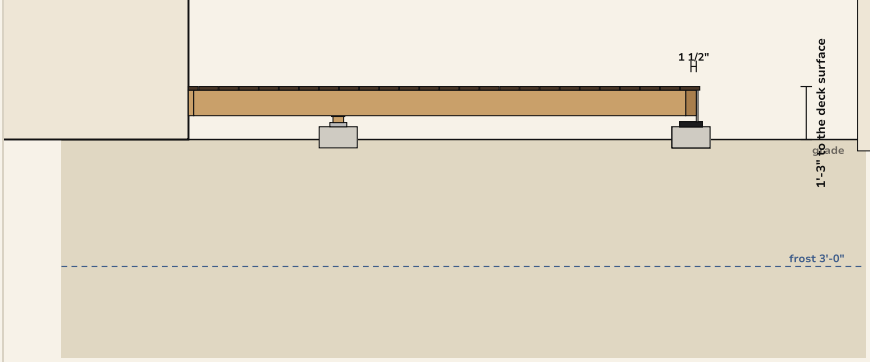
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

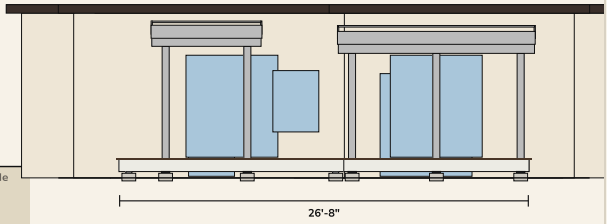
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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SECTION through zone M at 6'-0" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 1'-3" above grade

FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst

Posts 2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam on DP-50/50

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

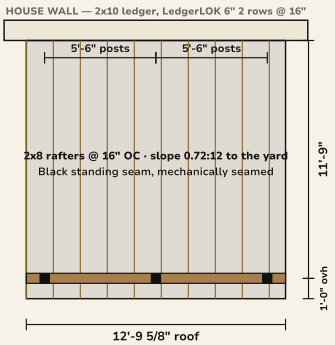
A-401 ROOF COVERS — PLAN + SECTION

Lean-to covers: ledger, rafters, beam, posts on the piers, clear heights, pitch, standing seam, Hardie, blinds

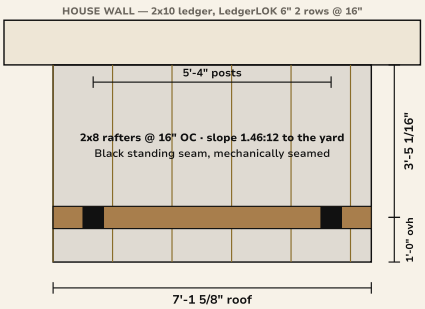
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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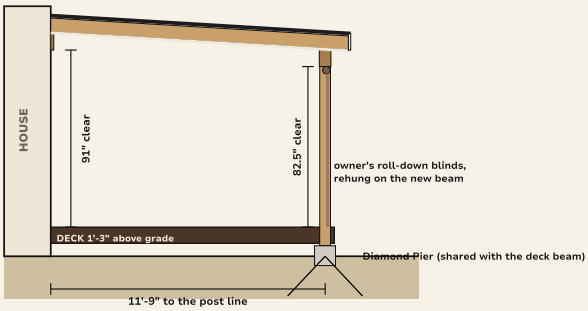
MAIN COVER — PLAN



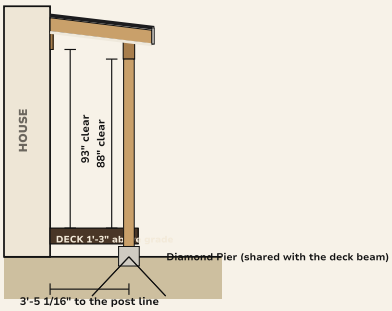
WALKWAY COVER — PLAN



MAIN COVER — SECTION



WALKWAY COVER — SECTION



Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 0.72:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · AC6Z caps
6x6 posts on ABA66Z bases on the Diamond Pier bolt · the engineer's uplift hold-down governs

Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 1.46:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · AC6Z caps
6x6 posts on ABA66Z bases on the Diamond Pier bolt · the engineer's uplift hold-down governs

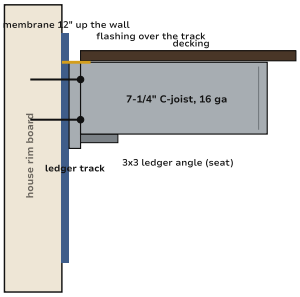
D-501 DETAILS — STEEL FRAME

Steel ledger · joist clips · boxed beam on the shared pier · Diamond Pier · deck edge · under the deck

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

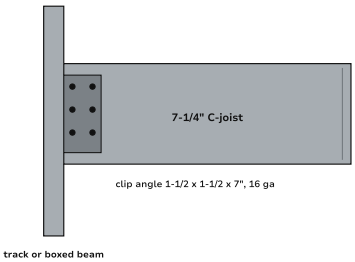
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1 STEEL LEDGER AT THE HOUSE



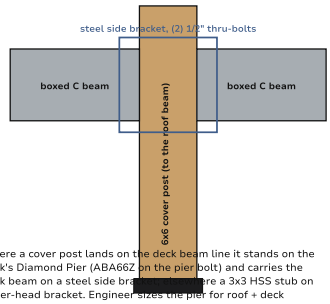
7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

2 JOIST TO TRACK / BEAM



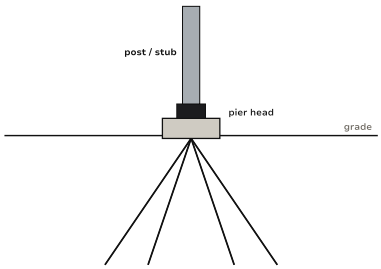
#12-14 self-drilling hex-washer screws, 4 per leg minimum (clip to joist web, clip to track / beam web) — the engineer's clip schedule governs. Cut ends and field holes cold-galvanized.

3 BOXED BEAM · SHARED PIER



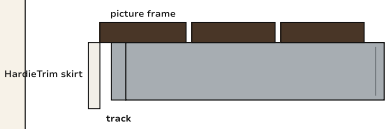
Where a cover post lands on the deck beam line it stands on the deck's Diamond Pier (ABA66Z on the pier bolt) and carries the deck beam on a steel side bracket; there a 3x3 HSS stub on a pier-head bracket. Engineer sizes the pier for roof + deck and the uplift hold-down.

4 DIAMOND PIER



DP-50/50 (4 pins) — no excavation; frost-protected by the pins (submit the evaluation report). 811 locate first.

5 DECK EDGE · HARDIE SKIRT



TimberTech Landmark American Walnut: hidden fasteners in the field, the picture frame face-fastened with color-matched plugs — every screw the maker's steel-frame screw for 14–18 ga. HardieTrim skirt on furring screwed to the track.

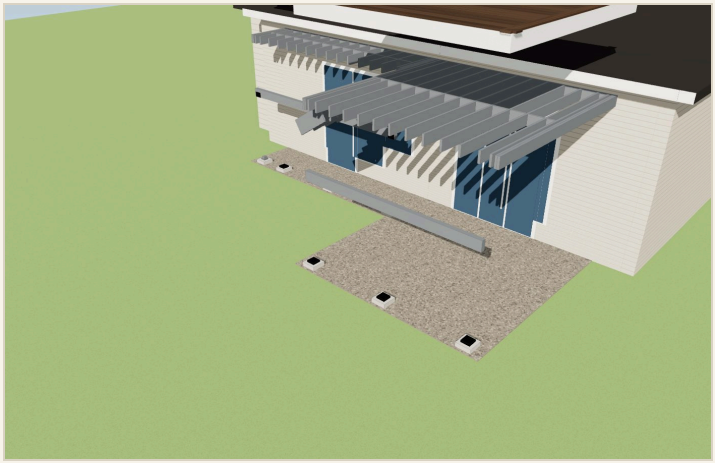
6 UNDER THE DECK · LOW END



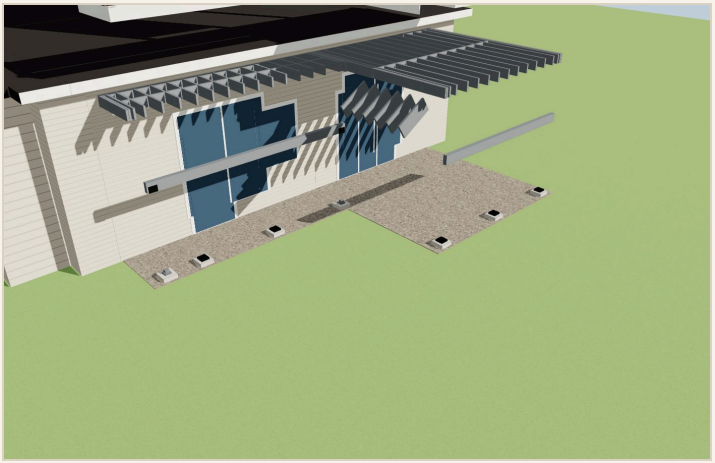
2" gravel over 6-mil poly, sloped away from the house

Walkway low point: 9" surface to grade leaves 0.75" under the frame — dig out / regrade to 2" min. under that run before the poly.

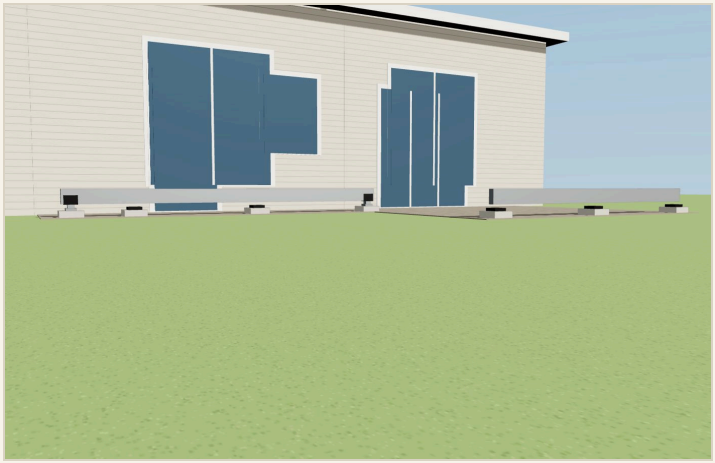
EXPLODED VIEWS



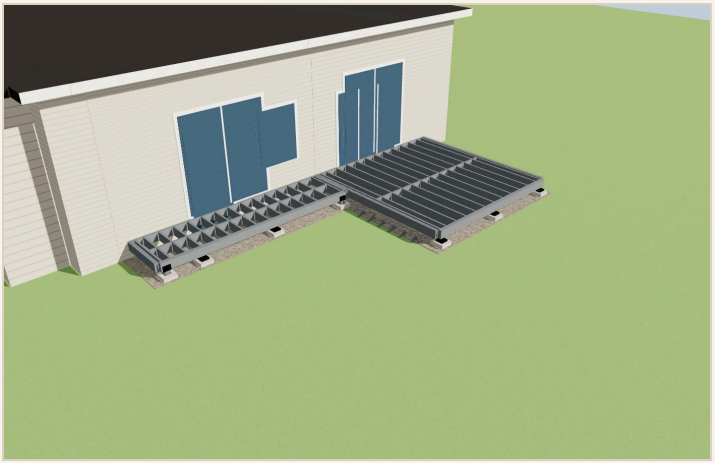
Exploded — every layer



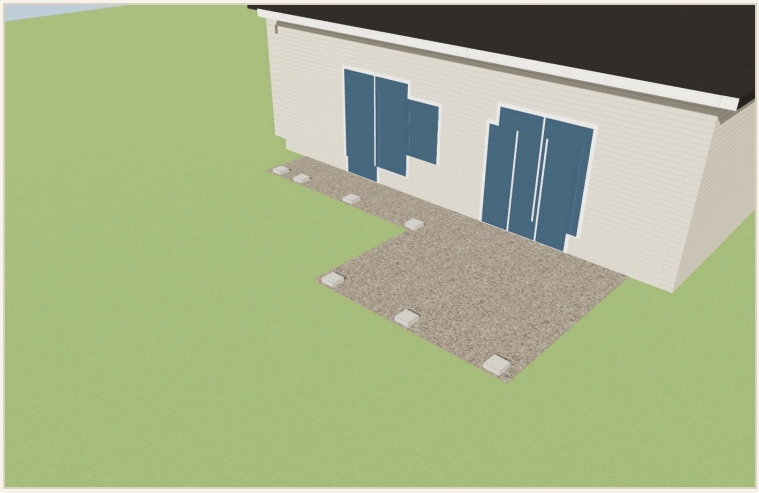
Exploded — from the open corner



Posts and beams on the footings



Frame only — joists, rims, beams, ledger



Step 1 · Footings — 7 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

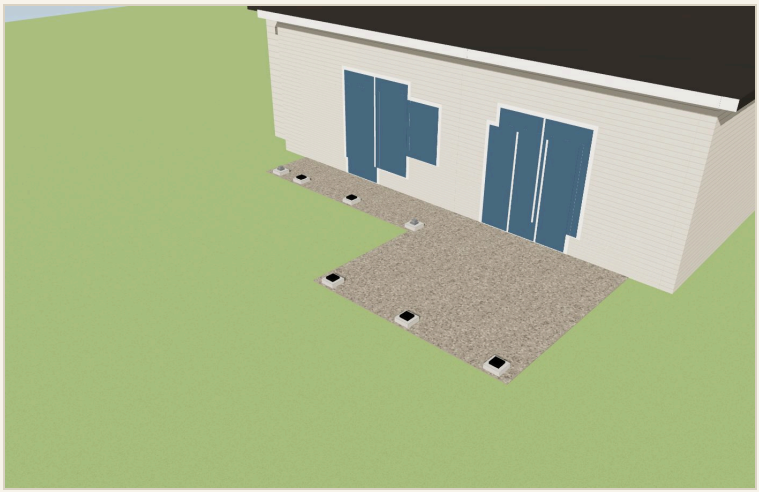
- (7) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (1) Upgrade to Diamond Pier DP-75/63 at the shared main cover posts (roof + deck on one pier)

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FLUSH boxed 7-1/4" steel C W at 3'-6 1/2" from the reference wall, posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16"; FLUSH boxed 7-1/4" steel C M at 11'-10 1/2" from the reference wall, posts at 15'-2" / 20'-8" / 26'-2".
- Set 7 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 #2 GC on ZMAX bases

T-401 · Step 2 · Posts

GOES IN

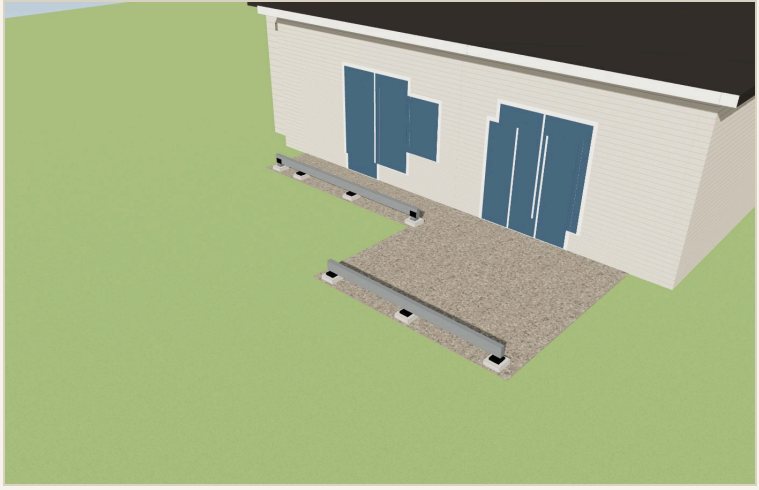
- (5) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)
- (3) Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover
- (2) Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): FLUSH boxed 7-1/4" steel C W posts 6" / FLUSH boxed 7-1/4" steel C M posts 6" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 4 · Beams — FLUSH boxed 7-1/4" steel C W (4 posts) · FLUSH boxed 7-1/4" steel C M (3 posts)

T-403 · Step 4 · Beams

GOES IN

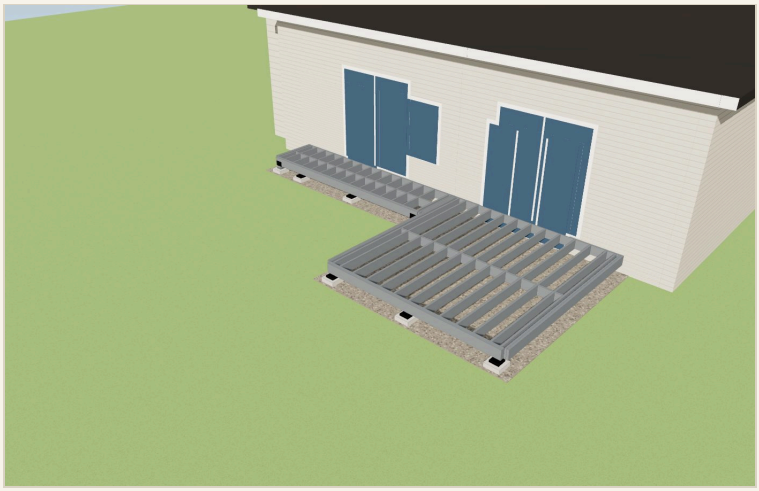
- (5) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)

HOW

- FLUSH boxed 7-1/4" steel C W: 14'-2" one piece, 4 posts @ 5'-9 3/16" max
- FLUSH boxed 7-1/4" steel C M: 11'-6" one piece, 3 posts @ 5'-6" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 5 · Frame — 2x8 joists at 12", rims, ledger, blocking over the beam

T-404 · Step 5 · Frame

GOES IN

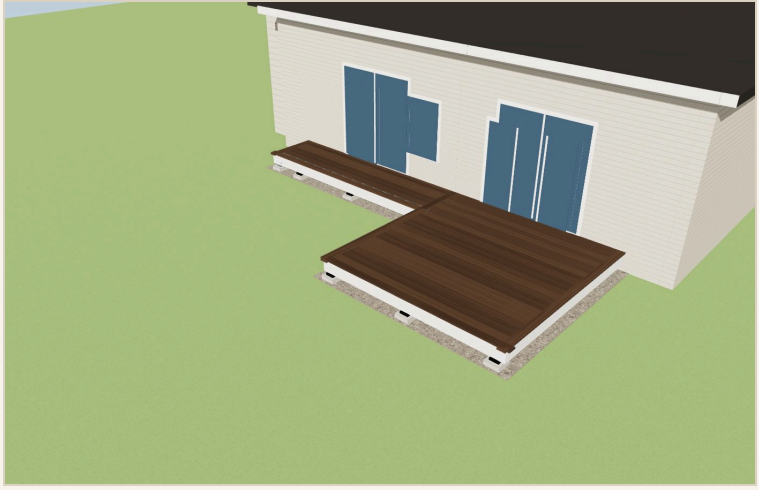
- (1) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (3) Aluminum L-flashing 10' (over ledger + side rims)

HOW

- 7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
- W: 13 joists @ 12" OC x 3'-3 1/2"; M: 11 joists @ 12" OC x 11'-7 1/2". Crown up, hangers at the ledger, both faces of the flush beam.
- Blocking one row over the beam, staggered.
- Tape every joist top.

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 6 · Decking — TimberTech Landmark American Walnut, American Walnut border, hidden fasteners

T-405 · Step 6 · Decking

GOES IN

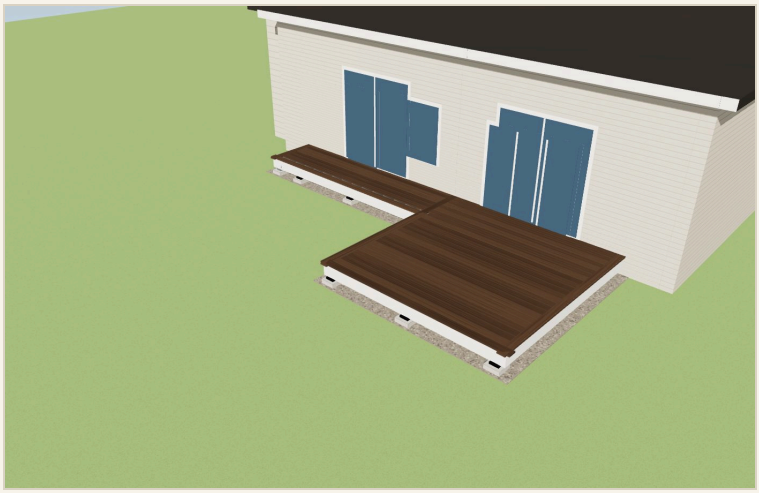
- (25) TimberTech Landmark American Walnut 1x6x12 Grooved
- (7) TimberTech Landmark American Walnut 1x6x16 Grooved
- (4) TimberTech Landmark American Walnut 1x6x16 Square Edge — borders & dividers
- (1) TimberTech Landmark American Walnut 1x6x20 Square Edge — borders & dividers
- (5) Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia)
- (1) Color-matched trim screws for HardieTrim, 100 ct (Arctic White)
- (1) ColorPlus touch-up kit, Arctic White

HOW

- CONCEALoc at every joist in every gap (437)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board
- Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · (no rail)

T-406 · Step 7 · Rail

GOES IN

HOW

CHECK

- —
- Cables tensioned / balusters under 4" clear
- End of construction



Step 8 · Your room

T-407 · Step 8 · Porch cover

GOES IN

- (3) 6x6x10 Douglas fir #1 post (cover) — Main cover
- (2) 2x10x16 #1 SYP — cover beam (2 ply) — Main cover
- (1) 2x10x16 #1 SYP — cover ledger on the house — Main cover
- (10) 2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover
- (3) 2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover
- (6) 7/16" OSB roof sheathing 4x8 — Main cover
- (170) Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 13.2' (Black, PVDF), 10 panels — Main cover
- (1) High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover
- (2) Standing seam eave / drip trim 10', color-matched — Main cover
- (3) Standing seam gable / rake trim 10', color-matched — Main cover
- (2) Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover
- (3) Foam + metal closure strips at eave and headwall, 10' — Main cover
- (1) Panel clips + pancake-head screws, box of 250 — Main cover
- (1) Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover

HOW

- Cover ledger on the house at the top of the pitch, LedgerLOK 2 rows; 6x6 Douglas fir posts on ABA66Z bases on the Diamond Piers (shared with the deck beam where they line up), (2)2x10 beam on the posts; 2x8 rafters @ 16" OC with hurricane ties and the overhang; 2x8 sub-fascia and OSB.
- High-temp self-adhered underlayment over every sheet, headwall flashing counterflashed into the house wall, eave trim, then the 24 ga snap-lock panels one piece eave to headwall — clips 18" OC on each seam, no face screws in the field; rake trim and closures last; snow retention bar clamped to the seams 12-18" above the low eave, the full width.

- Ceiling: HardieSoffit 4x10 cedarmill Arctic White panels 6" OC stainless ring-shank at every rafter; HardieTrim 4/4 x 7.25 smooth Arctic White fascia and rakes over the 2x8 sub-fascia, color-matched trim screws.

CHECK

- Ledger flashed and counterflashed — hose test before the ceiling closes
- Posts plumb over blocking, beam level, rafters tied
- Panels straight, seams engaged full length, no oil-canning; snow bar continuous
- Water sheds clear of the deck edge

3D MODEL



The interactive 3D model lives at the online version of this document.

Quantities from the drawings; the stamped engineering set governs structure. Quote requested — no pricing on this sheet.